

CHAIR FOR HEALTHCARE AND REHABILITATION ROBOTICS

TECHNICAL UNIVERSITY OF MUNICH

Report Submitted to Master-Praktikum: Artificial limbs - Design and control

Design of a Soft Foot for Lower-limb Prostheses

Nithin Balan, Sofía La Rota López



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Design of a Soft Foot for Lower-limb Prostheses

Entwurf eines weichen Fußes für Prothesen der unteren Gliedmaßen

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We confirm that this report is our own work and I have documented all sources and material used.

Munich, February 12, 2023 Nithin Balan, Sofía La Rota López

Abstract

Despite ongoing advances in the research of more lifelike lower-limb prosthetic devices that allow patients to recover a significant amount of their mobility, the majority of prosthetic feet used worldwide are significantly less sophisticated. State of the art bionic limbs and even the most sophisticated passive prosthetic feet are out of the reach and budget of rural populations in developing countries. The livelihood of this demographic is more likely to depend on physical activity, which includes extensive walking in rough terrain. In this report, we attempt to address this problem through the development of a soft prosthetic foot made out of compliant material that can both provide stability in rough terrain as well as a propulsive gait effect.

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1 Introduction

People from rural populations in developing nations often rely heavily on their physical fitness for their and their family's income, especially if they are dedicated to agriculture or similar, which makes lower-leg amputations particularly devastating. One such example is the case for Colombia, a country with an unfortunate history of warfare and a high percentage of disabled population due to lower-limb amputations. It is estimated that at least 4000 lower-limb prosthetic devices will be needed every year [1], either as a direct consequence of the war, from traffic accidents, snake bites or diabetes. According to statistics compiled from different non-profit institutions [2] and the National center for Historic Memory [3], there have been 11,500 recorded cases of victims killed or maimed from 1990 to 2017 due to these devices, although unofficial counts suggest upwards of 20,000. 40% of the victims were civilians, and 98% live in rural areas.

The most affected region is the northwestern province of Antioquia, with 17% of the cases. Antioquia's geography is characterized by the Andes mountain range and the valleys of the Cauca and Magdalena rivers. Rural towns in Colombia are built upon centennial architecture and connected by old colonial roads made of dirt and brick roads, traversed mainly via horseback riding, bicycle, motorcycle, and by foot. Some core Colombian cultural expressions, like folk dances and football, require ample foot dexterity. Not only are amputees at risk from losing most of their autonomy in terms of daily life activities and income, but face exclusion from social activities [1].

Despite recent governmental initiatives to repair the victims of the conflict, most prosthetic devices are either paid out of pocket by the patients themselves or handed out as donations from non-profit organizations, such as the Mahavir Kmina [4], which formerly accounted for a great part of the demand by creating hand-made prosthetic feet and legs based on the design of the Jaipur Foot Foundation [5] in India. Unfortunately, the artisanal nature of these prosthetic devices and lack of certifications led the foundation to an end in September of 2022. Briefly stated, there is an urgent need for the development in prosthetic feet that are both accessible and able to perform correctly for high-intensity walking in uneven terrain, and thus answer to the requirements of this particular demographic.

1.1 State of the Art

There are many prosthetic feet available on the market now for those who have had trans-tibial amputations, each with a slightly different feature set to improve control, comfort, and cosmetics. Generally speaking, there are three types of prosthetic feet: conventional feet, energy-storing and returning (ESR) feet, and the more advanced so-called "bionic" feet, which include active components. Although much research has been conducted in the development

of increasingly complex and lifelike active lower-limb prostheses, state-of-the art technology that could fully compensate for the level of physical activity needed by these demographics is prohibitively expensive, and so will not be considered for this context [6].

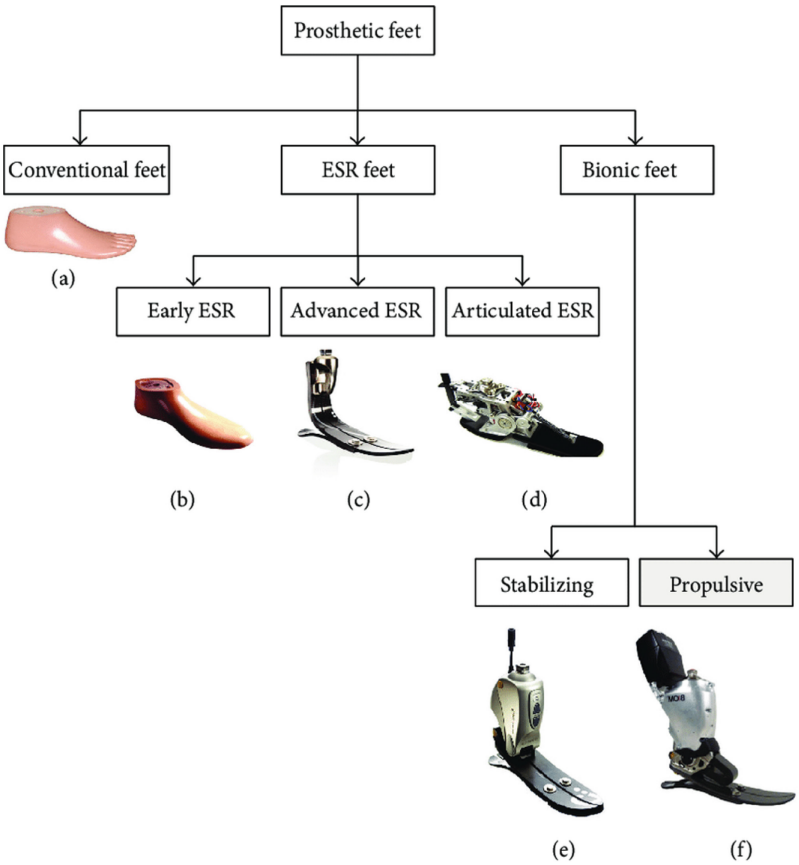


Figure 1.1: Categorization of today's prosthetic feet (a) the SACH foot, (b) the SAFE foot, (c) Ossur's Flex-Foot, (d) the CESR foot, (e) Ossur's Proprio Foot, and (f) iWalk's Powerfoot BiOM.[7]

Conventional Feet (CF): Conventional prosthetic feet aim to restore fundamental walking and enable amputees to do fundamental tasks, and are categorized by their degrees of freedom at the ankle level, being either completely rigid, uni-axial (allowing for plantar flexion and dorsiflexion) or multi-axial (further allowing for inversion and eversion). Although the higher mechanical complexity allows for a better adaptability to uneven terrains, it is also correlated with both a higher weight and cost, however. Modern multi-axial feet include the Seattle foot [8], which is directed for users with moderate levels of activity, however, and is not suitable for frequent contact with water. The most used prosthetic foot in use worldwide today is still completely rigid, which is recommended for people with even lower activity level, which may not truthfully correspond to he end-user's needs, but rather, to what they can afford [6] [9] [10].

Energy-Storing-and-Returning (ESR) feet: On the other hand, the prosthetic leg development has advanced greatly over the past few decades as a result of amputees' desires to move

more naturally, and in some cases, exercise. ESR feet are designed to store energy by loading a spring with the body weight and release it at the toe-off phase of the gait, creating a propulsive effect. ESR feet improve comfort levels and lower metabolic costs, although it can only partially imitate the normal ankle-foot complex[7]. The amputee's gait speed is still slower than normal walking and many studies have shown that ESR feet have not resolved the increased metabolic energy consumption that leads to early fatigue. In addition, natural walking patterns are still severely disturbed, leading to long-term medical conditions.

Jaipur Foot: There is an inherent trade-off between the cushioning, stabilizing and propulsive effects of soft feet, with both SACH and ESR feet having their advantages and drawbacks[6] [9] [10]. Historically, there have been some iterations done in order to compensate the disparities between economic cost and activity level required by the user. One notable example of this is the Jaipur foot [11], developed by Professor P.K. Sethi in the 1970s, which is a type of modified SACH foot in which the rigid heel is exchanged for a solid piece of vulcanized rubber acting like a flexible, multiaxial joint. This adds considerable range of motion that makes this prosthesis suitable for rough everyday use, including traversing swamped areas and even climbing trees, without added mechanical complexity. Although the non-standardized design allowed it to be quickly and cheaply manufactured by artisans, the Jaipur foot still suffers from several shortcomings: its weight and non-standardization, which leads to fluctuating product quality, reliability and durability [8].



Figure 1.2: Custom-made prosthesis designed with 3D computer models and 3D printing. [12]

3D-printed feet: Finally, we address a growing trend of customization made available by the advance in 3D printing. This technique enables new versions to be created quickly at a lower cost than conventional manufacturing methods. Most importantly for the project, they enable rapid prototype creation. Figure 1.2 shows a 3D-printed foot designed in Munich for a little girl who was born without her left foot.

2 Theoretical Framework

The human foot is the combination of a hind foot, a mid-foot, and a forefoot. 52 bones, 60 joints, 214 tendons, 38 muscles, and numerous blood vessels builds the human foot. The spinal cord, heart, and brain are all connected to the foot by these blood vessels. The top of the foot absorbs shock when walking and supports weight through a strong connection to the foot's bones[13]. Although different models exist, most studies show an average relative foot weight in the range from 1.5 to 2.0% of the nominal body weight BW of an adult caucasian male[14].

2.1 Walking Gait Biomechanics

Human gait can be loosely decomposed in three distinctive phases[15], as shown in figure 2.1. For a more detailed gait analysis, we suggest the work by Shamaei et al [16].

- Foot strike or heel strike: The foot receives an impact of roughly 1.2 times the body weight as it makes contact with the ground. The expected heel torsional stiffness is about about 3.53 Nm/degree during this phase.
- Dual-flexion: the entire sole and toe of the foot make contact with the ground (if it is flat) as the weight shifts from the back to the forefoot. The ground reactions are approximately 0.8 BW. During this phase, the heel exhibits dorsiflexion followed by dual-flexion, where it exhibits an elevated torsional stiffness of about 17.31 Nm/degree.
- Toe-Off: Finally, the terminal stance is characterized by plantarflexion and a propulsive effect where the foot and toes push off the floor with a force of about 1 BW. The expected heel torsional stiffness is about about 4.29 Nm/degree during this phase.

As for the heel stiffness from the inversion-eversion point of view, a linear regression of the data by the passive reactions measured by Fenfang et al[17] is considered, as seen in figure 2.2, and a torsional stiffness value of 11.05 Nm/degree is taken as reference.

2.2 Mechanical Model of the Foot

Although a thorough mechanical model of biological tissues such as tendons and ligaments should consider the viscoelastic properties[18] [19], we consider these to be out of the scope of this work. Instead, we will assume a static analysis and ideal elastic behavior under quasi-static deformation rates. Thus, materials could be thought of as ideal springs whose properties are defined by its geometry and Young's modulus.

Our simplified model of the foot considers only three degrees of freedom: A multi-axial heel capable of inversion-eversion and dorsiflexion-plantarflexion. The toe movement as a lumped

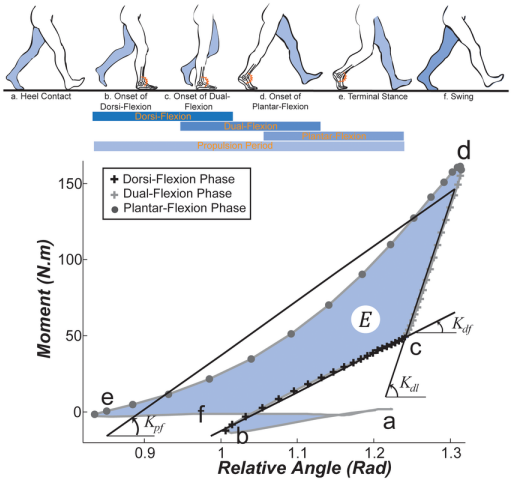


Figure 2.1: Ankle moment vs. relative angle curve for a representative subject walking at 1.75 m/s. [16]

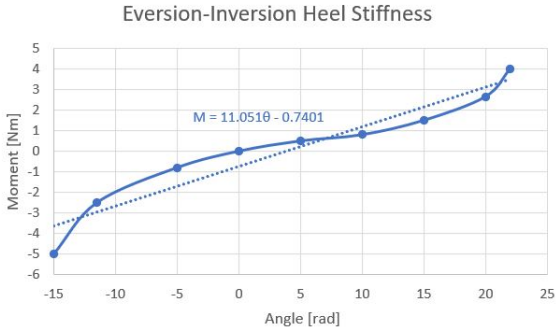


Figure 2.2: Heel inversion and eversion reaction torques and range of motion. [17]

mass is also taken into account. These three joints are modeled according to the pseudo-rigid model and taking the aforementioned stiffness values as reference, as illustrated by the schematic drawing in figure 2.3. A thorough explanation of Howell's pseudo-rigid theory and its principles can be found in [20].

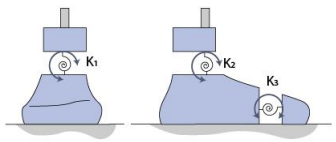


Figure 2.3: Pseudo-rigid schematic of a simplified human foot

3 Methodology

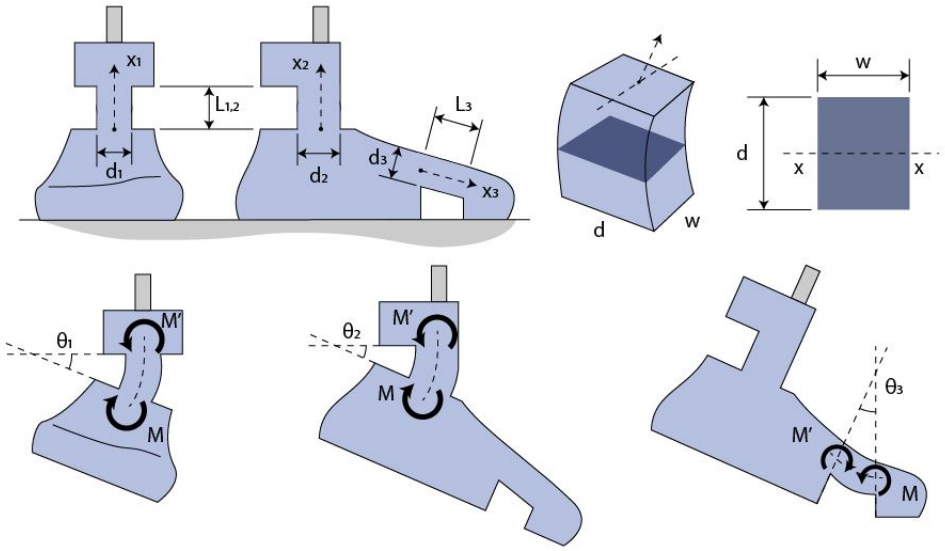


Figure 3.1: Graphical representation of the three feet joints as compliant segments.

In order to avoid unnecessary iterations between cycles of CAD design and finite-element-analysis (FEA), we "warm-started" our design process by finding approximated geometries for the compliant foot cross-sections according to the linear Euler Beam theory under the assumptions of constant bending moment M, a material with Young's modulus E, and a rectangular cross-section whose geometry can be detailed in figure 3.1. L, w, and d are the beam length, width and depth, respectively. Likewise, 1 represents the flexion given inversion/eversion movement, 2 represents dorsiflexion/plantarflexion, and 3 represents the toe joint. The known boundary conditions are the imposed foot deflection angles θ_{foot}:

$$M = E(wd^3/12L)\theta_{foot} \tag{3.1}$$

Finally, by substituting for the torsional stiffness constant $K = M/\theta$ into Eq. (3.1), the geometrical restrictions are imposed. K_{eq} represents the equivalent torsional stiffness from combining K_1 and K_3 in series, when neither dorsiflexion/plantarflexion or toe extension are unrestricted (when the foot is not in contact with the ground). The result would be the following indeterminate system of 7 equations with 14 unknowns:

$$K_1 = E(w_1 d_1^3/12L_1) \tag{3.2a}$$
$$K_2 = E(w_2 d_2^3/12L_2) \tag{3.2b}$$
$$K_3 = E(w_3 d_3^3/12L_3) \tag{3.2c}$$
$$1/K_{eq} = 1/K_1 + 1/K_3 \tag{3.2d}$$
$$w_1 = d_2 \tag{3.2e}$$
$$w_2 = d_1 \tag{3.2f}$$
$$L_1 = L_2 \tag{3.2g}$$

Given that the infinite number of solutions possible, some dimensional restrictions were imposed arbitrarily. This rough approximation allowed us to determine the feasibility of the material early on, given its Young's modulus. The equations were computed using the Young's modulus and the approximated maximum stress was calculated. Although the exact dimensions were not used in the final CAD, they provided a rough starting point to determine viability. This was programmed in Matlab.

3.1 Computer-Aided Design, Simulation and Optimization

The calculated values were used as a starting point for computer-aided design (CAD) using Solidworks. A static analysis of the design was performed using finite element analysys (FEA) in ANSYS Workbench.The goal of these simulations was to compare the maximum Von-Misses stresses against the proposed material yield stress with a safety factor of 3. All calculations included the option for large (nonlinear) deflections enabled and were done considered a body weight BW of 80 Kg (784.8 N). Four different simulations were performed, corresponding to different movements and loads. The maximum vertical gait reaction of 1.2 BW is used as a maximum value.

- Inversion-eversion (Inv./Ev.): The foot was rigidly restricted from the ankle while a ramped force of 941 N was applied. The force had vertical as well as lateral components, as it was applied of the surface of the foot sole at an angle of 45 degrees.
- Dorsiflexion-Plantarflexion (Dors./Plant.): The foot was rigidly restricted from the ankle while a ramped force of 941 N was applied. The force had vertical as well as lateral components, as it was applied of the surface of the foot sole at an angle of 45 degrees.
- Toe Joint movement (Toe): The foot was rigidly restricted from the ankle and heel while a ramped force of 941 N was applied directly upwards on the toe.
- Combined Heel and Toe Movement: The foot was rigidly restricted from the ankle while a ramped force of 941 N was applied directly upwards on the toe. Having the whole foot moving freely allowed the calculation of the equivalent stiffness of both the toe and heel joints combined.

Finally, the topological optimization option in ANSYS was used to further refine a satisfactory design. This allowed us to minimize prototype weight and printing time by removing structurally redundant material.

3.2 Computation of the Simulated Stiffness Values

Asides from the stress analysis, the simulation animation with the foot deformation was exported and used to track the angle as a function of the loads using Tracker Video Analysis Software. By using the known force inputs and the foot geometry, we calculated the deflection angles as a function of bending momentum and obtained the simulated stiffness constant by a linear regression of the plotted data. These results were compared to the theoretical ranges.

3.3 Qualitative Analysis

ANSYS Workbench was also used to simulate the effects of a normal gait cycle, as well as the effects of stepping on different obstacles. Several forces were applied to different parts of the foot sole and visually compared by subjecting the prototype to experimentally similar conditions. Different types of rigid obstacles were analyzed: large terrain perturbations similar to steps, medium-sized stones, and smaller-sized objects such as a bottle caps.

The experiment was carried out by having a person using the built-in support in order to load their entire body weight on the foot as it stepped on an obstacle. This person would have access to a nearby structure (like a wall or another person) in order to provide support for compensatory movements necessary for stabilization, as shown in figure 3.2. This person reported how difficult it was to maintain a stable position under those circumstances. Photographic evidence of the experiments was also analyzed with this in mind.



Figure 3.2: A person using the prototype foot and structure to step on a bottle cap.

4 Soft Foot Design

The design criteria for the foot are based on the context requirements. Most importantly, the foot has to be able to flex passively in a controlled manner without risking material failure under the loads characteristic of normal human gait and with characteristic joint stiffnesses similar to those of a real foot. It should also strive to be as light to a human foot or lighter.

Furthermore, we tried to find a balance between propulsion and stability while walking in rigid terrains. Both effects can be achieved by elastic sections of different thicknesses: thicker, stiffer sections behave like leaf springs while thinner sections and cavities allow the material to conform around perturbations and act as dampers.

Regarding aesthetics, static cosmesis to a real human foot is not considered. While it is important that the foot movement looks similar to how a real foot would move during normal gait, an in-depth study of the effects of this prosthetic device on a patient's gait is out of the scope of this project. However, we tried to maintain an aesthetic, if not lifelike appearance.

A final consideration includes foregoing mechanical complexity in favor of geometrical complexity. As mentioned by Howell [20] in his theory, multi-body mechanisms imply rising costs, error propagation and assembly uncertainties, between others. The problem with traditional multi-axis prosthetic feet is the rising cost and weight due to increased mechanical complexity. Meanwhile, conventional problems regarding geometrical complexity are easily bypassed by considering additive manufacturing methods such as 3D printing.

4.1 Material Considerations

A rapid prototype of the foot was 3D-printed in thermoplastic polyurethane (TPU) Flex Hard with an 80% infill. The following material properties were taken as reference values:

- Young's Modulus: 40 MPa
- Density: 1.2 g/cm^3
- Stress at break: 29 MPa
- Maximum Stress (with a safety factor of 3): 9.67 MPa

4.2 Design Progression

The prototype foot design is heavily inspired in the geometry of ESR feet models. Its main geometric features are the thinner cross-sections where it is expected to bend, with rounded radii to avoid stress concentrators. A hollow cavity in the heel is added to allow the sole

of the foot to deform and absorb the impact from heel-strike. Figure 4.1 shows the design progression: (a) The first iteration included rough measurements similar to one of the authors' feet and compliant sections with loosely based on the first numerical approximation dimensions. However, preliminary simulations showed excessive inversion-eversion movement amplitude. (b) Addressing these concerns, a second version includes a physical guard to limit this rotation amplitude to about 12 degrees in either direction, and the cross-section is reinforced as well. The sole was re-designed to allow for local deflections in the hopes that it would be better able to conform around small pebble-like obstacles instead of addressing these perturbation at the heel. (c,d) Further refinement of the design after topological optimization, in which structurally redundant material from the center of the toe region is removed. Provisions for attachments to a testing structure are included. (e) CAD assembly of the foot rigidly attached support structure designed for it using standard components. (f) Finished, printed prototype. White dots were painted to allow for easier visualization of the foot deformation.



Figure 4.1: Custom-made prosthesis designed with 3D computer models and 3D printing. [12]

5 Results

The foot prototype was successfully printed and assembled onto its support with manual tools. The simulated weight according to Solidworks (assuming a completely solid foot) gave us an upper weight bound of 0.99 kg weight, whereas the marker shown by the printer slicer indicated an estimated weight of 972.69 g. At any rate, even a completely solid design succeeds in being 28 +/- 20% lighter than the expected 1.2-1.6 Kg of an 80 Kg patient's foot.

5.1 First Numerical Approximation

A Matlab program was written and used to iterate solutions of the system of equations 3.2. Figure 5.1 shows the results of one acceptable numerical approximation, which shows reasonable dimensions. The numbers in bold were predetermined constants. The CAD models generated afterwards have sections whose dimensions are loosely based on these results.

Joint	D	W	L	Torsional Stiffness Constant	
Units	cm	cm	cm	N*m/rad	N*m/degrees
Heel (Inv./Ev.)	3.72	4.67	**2.00**	**400**	7
Heel (Dors./Plant.)	4.67	3.72	**2.00**	**630**	11
Toe Joint	**3.00**	8.89	**2.00**	**400**	7
Combined Heel & Joint	---	---	---	315	3.5

Figure 5.1: Approximate dimensions and torsional stiffness constants for the joint design.

5.2 Simulated Torsional Stiffness Values

Figure 5.2 shows the results of the simulations as graphs for the four different movements tested. The left column shows the angular movement progression as the forces were applied. The right column shows the angular deflection versus equivalent bending torque. A linear regression of these graphs provided the simulated torsional stiffness constants for the joints. One interesting observation is how the toe and heel movement combine to generate a lower equivalent stiffness, as they are effectively acting as two torsional spring in series during the toe-off stage of gait. Given geometrical constraints, which would imply the toe joint would being restricted during dual-flexion, only the heel joint can move when the whole foot is on the floor, effectively making it stiffer.

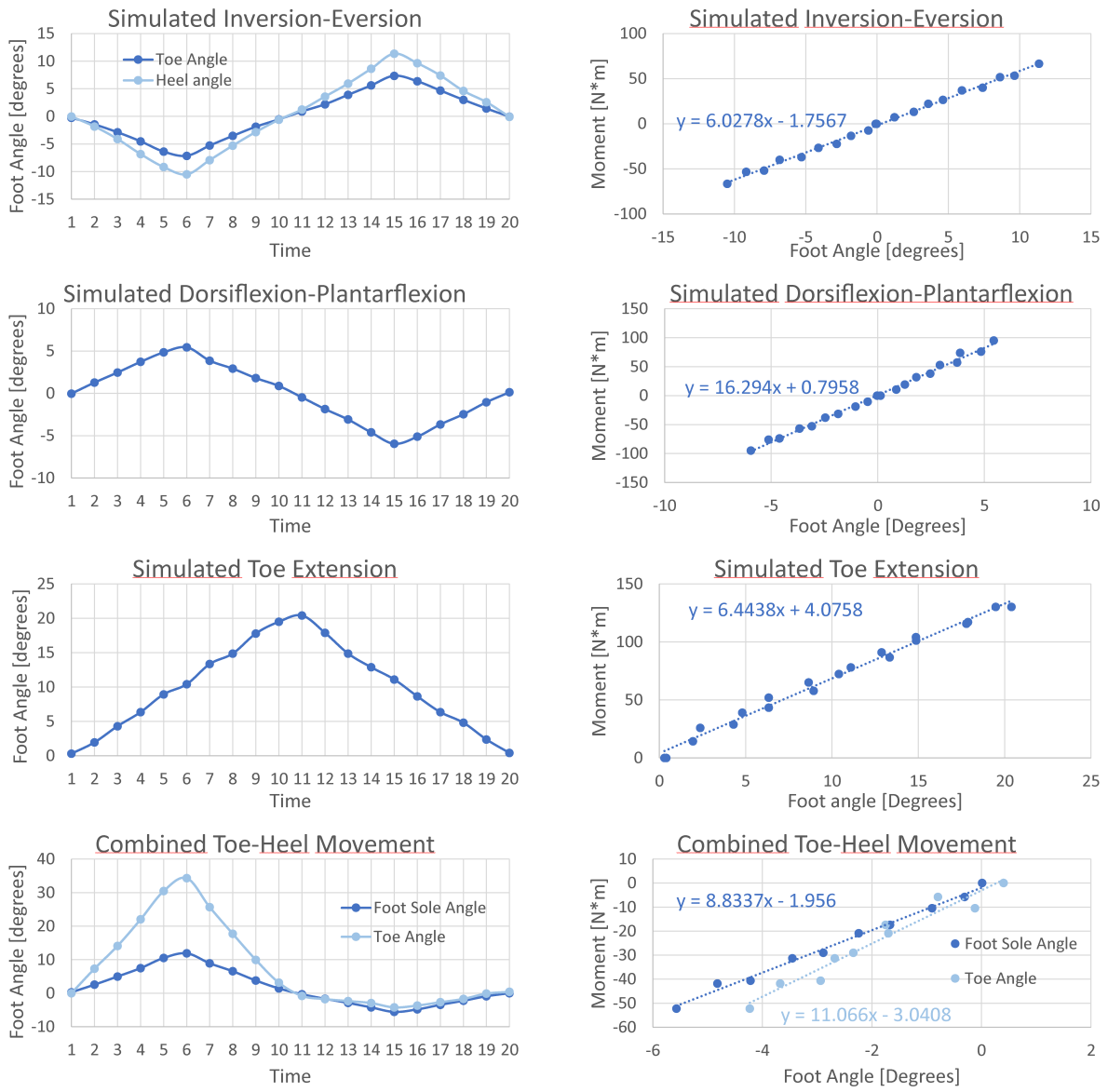


Figure 5.2: Angular deflection of the foot for different configurations.

Figure 5.3 compares the results for the simulated and theoretical values of the foot joints, and illustrates the maximum on Misses stress reported by ANSYS. A comparison between the simulated torsional stiffness values shows great similarity with biometric reference values, albeit on the stiffer end of the spectrum for the dorsiflexion-plantarflexion values. The inversion-eversion joint is 45.5% softer than the reference value, but this did not generate noticeable problems or instability, as the angles were restricted to less than 12 degrees to either side and did not generate buckling when tested. Of the four configurations simulated, only the combined heel and toe simulations barely exceeded the safety stress limit of 9.67 MPa.

Joint	Stiffness Reference	Simulated	Movement Range	Maximum Von Misses Stress
Units	[N*m/degr]	[N*m/degr]	[degr]	[MPa]
Heel (Inv./Ev.)	11.05 [19]	6.02	0 - 11.35 (Inv) 0 - 10.5 (Ev.)	6.42
Heel (Dors./Plant.)	3.53 – 17.31 [15]	16.29	0 - 5.95 (Dors.) 0 - 5.44 (Plant.)	4.66
Toe Joint	---	6.44	0 - 20.40	9.32
Combined Heel & Toe Joints	3.53 – 17.31 [15]	8.83	0 - 34.31 (Toe) 0 - 4.23 (Heel)	10.39

Figure 5.3: Numerical results of the simulations, with comparison to theoretical heel torsional stiffness values.

5.3 Simulated Gait Cycle

Figure 5.4 illustrates a visualization of three stages of the gait cycle. A force equivalent to 1.2 BW was applied on the heel for the foot-strike. Visually, the heel cavity deforms at this stage as it absorbs the impact without causing plantarflexion on the heel. Then, a distributed force of 0.8 BW was applied on the sole and toe joints for the dual-flexion phase. No major deformations occur. Finally, 1 BW was applied on the toe joint for the toe-off. We can see the heel and toe joints bending in a similar way to normal human gait, although the dorsiflexion is more pronounced. Superficially, the foot seems to move similarly to a regular foot, even if all the joints are passive. Additionally, this deformation causes a palpable propulsive effect.

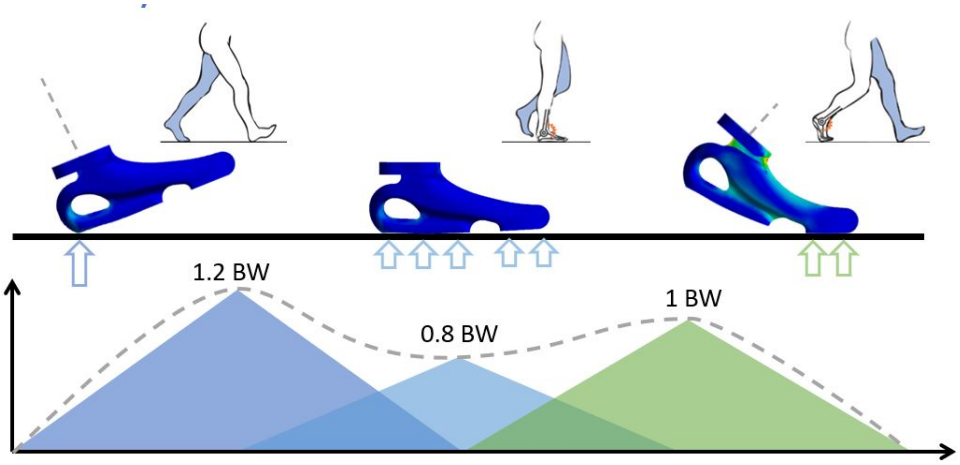


Figure 5.4: Numerical results of the simulations, with comparison to theoretical heel torsional stiffness values.

5.4 Behavior in Rough Terrain

Figure 5.5 shows close-ups of the prototype being used to step on a medium-sized stone. The toe joint flexed under the weight of a person, allowing the foot to maintain contact with the floor, a mild ankle angle, and requiring little balancing effort from the volunteer on top. The leftmost image clearly shows the foot sole conforming around the surface of the stone, which exerts a somewhat distributed force on it. However, this stability was compromised when the stone was switched to a smaller bottle cap, as shown in figure 5.6, which has an effect more akin to a point force. The sole and toe surfaces are too locally rigid, and a large portion of the foot is left in the air. The volunteer stepping on the prototype required a considerable effort to remain static. Believe this problem to be inherent to the foot's design, as simulations with similar loads generated visually similar deformations. However, it is worth noting that the person stepping on the prototype weighted significantly less than 80 kg, which may have contributed towards a less compliant sensation.



Figure 5.5: Prototype behavior under a medium-sized stone.

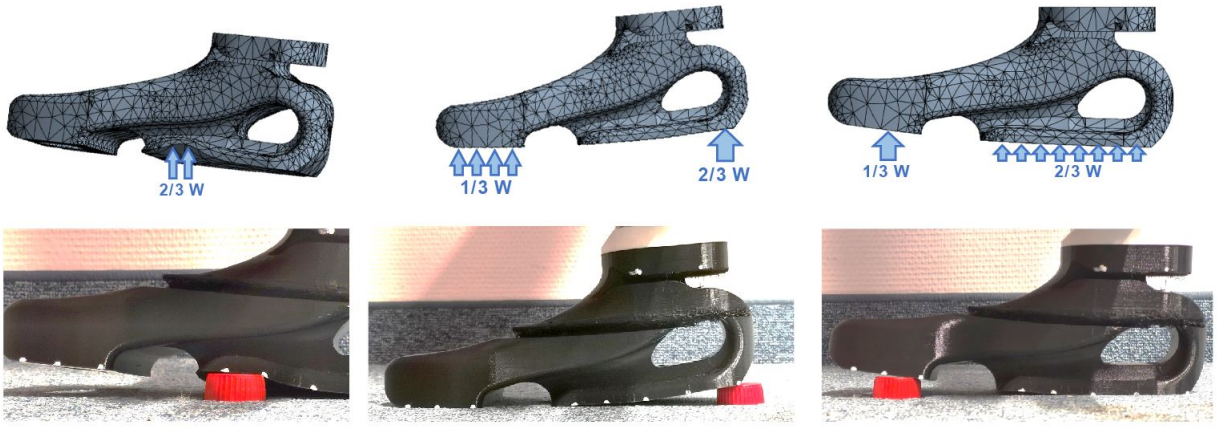


Figure 5.6: Visual similarity of the simulation and pictures of the prototype foot stepping on a small, firm object.

6 Conclusions

In general, we consider our soft foot design to be a successful step in the direction of addressing the current problems that many lower-limb amputees continue to face every day. The design generates a push-off effect at toe-off and succeeds at emulating a multi-axial foot without the disadvantages of added weight or mechanical complexity from a more traditional approach.

However, more work needs to be done in refining the soft foot design, especially when it comes to its behavior when facing small, rigid obstacles. One possible solution could be to design certain sections to be more sponge-like in structure, with thin sections and cavities that could mold themselves around obstacles better. The possibility of a forked toe section, which is present in many prosthetic feet in the market, could also be studied.

In any case, further testing is needed before any real conclusions can be drawn in terms of actual gait analysis and patient comfort. It would also be important to corroborate the simulation results with a quantitative experimental validation. For example, the methodology followed by professor Piazza's soft foot [21] could be followed.

Finally, it is important to consider the long-term dynamic effects of the structural integrity of the foot as well. Aspects such as material viscoelasticity, hysteresis, creep and fatigue play a major role in material selection as the foot is subjected to several load cycles and normal wear and tear.

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